

Woojung Son

✉ w.son@ufl.edu
📍 Gainesville, FL 32611

Education

University of Florida

M.S. in Agricultural and Biological Engineering
GPA: 3.89/4.0

Gainesville, FL
May 2025 – May 2027 (Expected)

Chonnam National University

B.S. in Bio-Industrial Mechanical Engineering
GPA: 4.32/4.50 | *Summa Cum Laude*

Gwangju, South Korea
Feb. 2021 – Feb. 2025

Research Interests

Spatial Multimodal LLMs, 3D Scene Representation, Vision-Language-Action Models

Experience

University of Florida

Graduate Research Assistant

USDA Robotics Strawberry Harvesting Project | Advisor: Prof. Won Suk Lee | In collaboration with Purdue University

Gainesville, FL

May 2025 – May 2027

- Collected the first real-world 6D pose ground truth dataset of strawberries in agricultural fields and constructed a scene-level realistic synthetic dataset using NVIDIA Isaac Sim.
- Evaluated baseline 6D pose estimation models and analyzed the sim-to-real gap in agricultural field conditions.

Chonnam National University

Teaching Assistant, Computer Programming

Assisted undergraduate students in C++, MATLAB, and Simulink.

Gwangju, South Korea

Spring 2024

Publications

- [1] **Woojung Son**, Won Suk Lee, Zijing Huang, Daeun Choi, Catia Silva, Yu She, Yan Gu, “From Simulation to the Real-World: An In-Field 6D Pose Dataset and Baseline for Robotic Strawberry Harvesting”, *arXiv preprint*, 2026.  

Presentations

- [1] **Woojung Son** et al., “Geometry-Grounded Transformer-Based Monocular 6D Pose Estimation for Robotic Strawberry Harvesting via Sim-to-Real Transfer”, *American Society of Agricultural and Biological Engineers (ASABE) Annual International Meeting*, Indianapolis, IN, 2026. (Oral)

Honors and Awards

Graduate Assistantship (Full Funding), University of Florida

2025 – 2027

1st Place, AKABFE Student Paper Competition, ASABE Annual International Meeting

2026

Skills

Programming: Python, C++

Libraries: PyTorch, OpenCV, Open3D, Hugging Face Transformers

Tools: Git, SLURM, L^AT_EX, NVIDIA Isaac Sim, COLMAP